

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV
COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

- (a) Clearly interpret the scenario and explain the cause of low contrast.
- (b) Identify all key issues affecting image visibility.
- (c) Apply an appropriate enhancement technique to address the problem.
- (d) Justify your choice with logical reasoning.
- (e) Present your explanation in a clear and structured manner.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

- (a) Interpret the scenario and explain the impact of uneven illumination.
- (b) Identify key factors causing intensity variation.
- (c) Apply a suitable enhancement method to normalize illumination.
- (d) Justify your selected approach with reasoning.
- (e) Provide a clear and well-structured explanation.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

- (a) Explain the scenario and characteristics of this noise.
- (b) Identify key issues impacting image quality.
- (c) Apply an appropriate filtering technique.
- (d) Justify why this method is more suitable than alternatives.
- (e) Present your response clearly and logically.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

- (a) Interpret the scenario and explain why edges are blurred.
- (b) Identify key issues related to smoothing and edge loss.
- (c) Apply an appropriate technique to restore edges.
- (d) Justify your decision with clear reasoning.
- (e) Structure your explanation clearly.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

- (a) Interpret the scenario and describe high-frequency noise.
- (b) Identify the key issues affecting image clarity.
- (c) Apply an appropriate frequency domain filtering method.
- (d) Justify the choice of filter with reasoning.
- (e) Present your answer in a clear and organized manner.

6. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

7. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

8. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

- (a) Interpret the scenario and explain segmentation challenges.
- (b) Identify key issues such as intensity overlap.
- (c) Apply an appropriate advanced segmentation method.
- (d) Justify how the method handles complexity.
- (e) Provide a structured and clear explanation.

9. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- (a) Interpret the scenario and explain why edges are broken.
- (b) Identify key issues affecting boundary continuity.
- (c) Apply a suitable method to improve edge connectivity.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

10. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

- (a) Interpret the scenario and define the combined requirement.
- (b) Identify key issues involving noise and edge preservation.
- (c) Apply an appropriate sequence of techniques.
- (d) Justify the order and selection with strong reasoning.
- (e) Present your response in a clear and structured format.

11. Low Brightness Image

An image captured in low light appears very dark with poor visibility.

- (a) Interpret the scenario and explain the cause of low brightness.
- (b) Identify key issues affecting image visibility.
- (c) Apply an appropriate enhancement technique.
- (d) Justify your choice with reasoning.
- (e) Present your explanation clearly.

12. Overexposed Image

An image appears excessively bright, losing important details.

- (a) Interpret the scenario and explain overexposure.
- (b) Identify key issues affecting intensity representation.
- (c) Apply a suitable correction technique.
- (d) Justify your choice logically.
- (e) Present your explanation in a structured manner.

13. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.

- (a) Interpret the scenario and describe Gaussian noise.
- (b) Identify key issues affecting quality.
- (c) Apply an appropriate smoothing filter.
- (d) Justify your method over alternatives.
- (e) Present your explanation clearly.

14. Speckle Noise

An image obtained from ultrasound imaging shows speckle noise.

- (a) Interpret the scenario and explain speckle noise characteristics.
- (b) Identify key issues affecting clarity.
- (c) Apply a suitable filtering technique.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

15. Loss of Fine Details

An image loses fine details after processing.

- (a) Interpret the scenario and explain detail loss.
- (b) Identify key issues related to filtering.

- (c) Apply a suitable enhancement technique.
- (d) Justify your method with reasoning.
- (e) Present your explanation clearly.

16. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.

- (a) Interpret the scenario and explain the need for edge enhancement.
- (b) Identify key issues in detecting edges.
- (c) Apply an appropriate method.
- (d) Justify your choice logically.
- (e) Present your explanation clearly.

17. Background Noise Removal

An image contains background noise affecting segmentation accuracy.

- (a) Interpret the scenario and explain noise impact.
- (b) Identify key issues affecting segmentation.
- (c) Apply a suitable preprocessing method.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

18. Threshold Selection Problem

An image requires segmentation, but selecting a threshold is difficult.

- (a) Interpret the scenario and explain the challenge.
- (b) Identify key issues in threshold selection.
- (c) Apply a suitable thresholding method.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

19. Texture-Based Segmentation

An image contains regions distinguishable by texture rather than intensity.

- (a) Interpret the scenario and explain segmentation challenges.
- (b) Identify key issues with intensity-based methods.
- (c) Apply an appropriate segmentation approach.
- (d) Justify your method logically.
- (e) Present your explanation clearly.

20. Multiple Object Detection

An image contains multiple objects with similar intensity values.

- (a) Interpret the scenario and explain segmentation difficulty.
- (b) Identify key issues in distinguishing objects.
- (c) Apply a suitable segmentation technique.
- (d) Justify your choice logically.
- (e) Present your explanation clearly.

21. Edge Noise Problem

Edges detected are affected by noise.

- (a) Interpret the scenario and explain the issue.
- (b) Identify key challenges in edge detection.
- (c) Apply a suitable solution.
- (d) Justify your method logically.
- (e) Present your explanation clearly.

22. Region Merging Requirement

An image contains fragmented regions that need merging.

- (a) Interpret the scenario and explain the issue.
- (b) Identify key challenges in region segmentation.
- (c) Apply an appropriate technique.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

23. Over-Segmentation

An image is divided into too many regions after segmentation.

- (a) Interpret the scenario and explain over-segmentation.
- (b) Identify key issues affecting segmentation.
- (c) Apply a suitable correction method.
- (d) Justify your choice logically.
- (e) Present your explanation clearly.

24. Under-Segmentation

Important regions are merged incorrectly during segmentation.

- (a) Interpret the scenario and explain under-segmentation.
- (b) Identify key issues affecting separation.
- (c) Apply a suitable solution.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

25. Boundary Refinement

Detected boundaries are rough and inaccurate.

- (a) Interpret the scenario and explain the issue.
- (b) Identify key challenges in boundary detection.
- (c) Apply a suitable refinement method.
- (d) Justify your choice logically.
- (e) Present your explanation clearly.

26. Frequency-Based Enhancement

An image requires enhancement using frequency domain methods.

- (a) Interpret the scenario and explain frequency-based processing.

- (b) Identify key issues in spatial domain methods.
- (c) Apply an appropriate frequency domain technique.
- (d) Justify your choice logically.
- (e) Present your explanation clearly.

27. Mixed Noise Scenario

An image contains both Gaussian and impulse noise.

- (a) Interpret the scenario and explain challenges.
- (b) Identify key issues affecting filtering.
- (c) Apply a suitable combined approach.
- (d) Justify your method logically.
- (e) Present your explanation clearly.

28. Edge Preservation Filtering

A filtering method is required to remove noise but preserve edges.

- (a) Interpret the scenario and explain the requirement.
- (b) Identify key issues in filtering.
- (c) Apply a suitable technique.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

29. Region Growing Failure

Region growing fails due to incorrect seed selection.

- (a) Interpret the scenario and explain the issue.
- (b) Identify key challenges in region growing.
- (c) Apply a suitable correction.
- (d) Justify your method logically.
- (e) Present your explanation clearly.

30. Industrial Inspection Enhancement

An industrial system requires detecting defects in noisy images.

- (a) Interpret the scenario and define requirements.
- (b) Identify key issues in detection.
- (c) Apply an appropriate enhancement and segmentation pipeline.
- (d) Justify your sequence logically.
- (e) Present your explanation clearly.

Code of Conduct:

I Bhavya PS(192524017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with	Makes reasonable decisions	Makes weak decisions with limited justification	No clear decisions made or

		strong rationale	with some justification		rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Low Contrast Image

(a) Interpret the scenario and explain the cause of low contrast

A grayscale image appears dull because the intensity values of the objects and the background are very similar. Since there is only a small difference between the darkest and brightest pixels, the image lacks visual distinction. This usually occurs due to poor lighting conditions, improper camera exposure, fog, haze, or limited dynamic range of the imaging sensor.

(b) Identify all key issues affecting image visibility

- Small intensity difference between foreground and background.
- Poor visibility of important objects.
- Loss of fine details.
- Reduced edge sharpness.
- Difficulty in image analysis, segmentation, and object recognition.

(c) Apply an appropriate enhancement technique

Technique: Histogram Equalization (or Adaptive Histogram Equalization - CLAHE)

Steps:

1. Read the grayscale image.
2. Compute the image histogram.
3. Redistribute pixel intensities over the full intensity range.
4. Generate an enhanced image with improved contrast.

(d) Justify your choice

Histogram Equalization spreads the intensity values across the entire grayscale range (0–255). This increases contrast and makes objects more distinguishable from the background. CLAHE performs even better because it enhances local contrast while preventing excessive brightness amplification.

(e) Structured Explanation

Problem: Low contrast image

Cause: Similar intensity values

Technique: Histogram Equalization / CLAHE

Result: Improved visibility, better edge distinction, enhanced image quality.

2. Uneven Illumination

(a) Interpret the scenario

The image contains bright and dark regions because lighting is not uniformly distributed across the scene. Some areas receive more illumination while others remain underexposed.

(b) Key factors causing intensity variation

- Uneven lighting sources.
- Shadows.
- Camera exposure variation.
- Surface reflections.
- Non-uniform illumination environment.

(c) Apply a suitable enhancement method

Technique: Adaptive Histogram Equalization (CLAHE)

Alternative methods:

- Homomorphic Filtering
- Background subtraction

(d) Justification

CLAHE improves local contrast independently in different image regions. Instead of treating the entire image equally, it enhances dark regions without over-enhancing already bright areas. Therefore, illumination becomes more balanced.

(e) Structured Explanation

Problem: Non-uniform brightness

Cause: Uneven lighting

Method: CLAHE

Result: Balanced illumination with improved local contrast.

3. Salt-and-Pepper Noise

(a) Explain the scenario

Salt-and-pepper noise appears as randomly distributed white and black pixels caused by transmission errors, faulty sensors, or scanning artifacts.

(b) Key issues

- Random bright and dark pixels.
- Poor readability.
- Loss of text clarity.
- Difficulty in OCR and segmentation.

(c) Apply an appropriate filtering technique

Technique: Median Filter

Steps:

1. Select a neighborhood window.
2. Sort neighboring pixel values.
3. Replace the center pixel with the median value.

(d) Justification

Unlike averaging filters, the Median Filter removes impulse noise without blurring image edges. Since it preserves boundaries while eliminating isolated noisy pixels, it is the most suitable filter for salt-and-pepper noise.

(e) Structured Explanation

Problem: Salt-and-pepper noise

Technique: Median Filtering

Reason: Removes impulse noise while preserving edges.

4. Blurred Edges

(a) Interpret the scenario

During noise removal, excessive smoothing has removed high-frequency information, causing object boundaries to become blurred.

(b) Key issues

- Loss of edge information.
- Reduced image sharpness.
- Difficulty in object detection.

- Reduced segmentation accuracy.

(c) Apply an appropriate technique

Technique: Unsharp Masking or Laplacian Sharpening

(d) Justification

Sharpening techniques enhance high-frequency components responsible for edges. Unsharp masking restores edge clarity while maintaining overall image quality.

(e) Structured Explanation

Problem: Blurred edges

Cause: Excessive smoothing

Technique: Unsharp Masking / Laplacian

Result: Restored sharp edges and improved detail.

5. High-Frequency Noise

(a) Interpret the scenario

A satellite image contains fine-grained random noise represented by rapid intensity changes. These unwanted high-frequency components reduce feature visibility.

(b) Key issues

- Fine-grained noise.
- Reduced clarity.
- Poor feature extraction.
- Difficulty identifying terrain details.

(c) Apply a frequency-domain filtering method

Technique: Low-Pass Filter (Gaussian Low-Pass Filter)

Steps:

1. Convert image to frequency domain using FFT.
2. Apply Gaussian Low-Pass Filter.
3. Perform inverse FFT.

(d) Justification

High-frequency noise mainly exists in the high-frequency region of the frequency spectrum. A Gaussian Low-Pass Filter suppresses these components smoothly without introducing ringing artifacts, preserving important image structures better than an Ideal Low-Pass Filter.

(e) Structured Explanation

Problem: High-frequency noise

Method: Gaussian Low-Pass Filter

Reason: Removes noise while preserving image quality.

6. Object Boundary Detection

(a) Interpret the scenario

Medical images require accurate detection of tissue boundaries to identify organs, tumors, or abnormal structures.

(b) Key challenges

- Low contrast.
- Image noise.
- Weak boundaries.
- Similar tissue intensities.

(c) Apply an edge detection method

Technique: Canny Edge Detector

Steps:

1. Gaussian smoothing.
2. Gradient calculation.
3. Non-maximum suppression.
4. Double thresholding.
5. Edge tracking.

(d) Justification

The Canny Edge Detector provides accurate, thin, and continuous edges with reduced false detections. It performs better than Sobel or Prewitt because it includes noise reduction and hysteresis thresholding.

(e) Structured Explanation

Problem: Detect tissue boundaries

Method: Canny Edge Detection

Result: Accurate and continuous boundary detection.

7. Simple Segmentation

(a) Interpret the scenario

Objects have intensity values clearly different from the background, making separation straightforward.

(b) Key features

- High contrast.
- Distinct intensity levels.
- Uniform background.
- Minimal overlap.

(c) Apply segmentation technique

Technique: Thresholding (Global Thresholding)

Steps:

1. Select threshold value.
2. Pixels above threshold → Object.
3. Pixels below threshold → Background.

(d) Justification

Thresholding is computationally simple and highly effective when object and background intensities are clearly separated.

(e) Structured Explanation

Problem: Simple object separation

Technique: Thresholding

Result: Fast and accurate segmentation.

8. Complex Segmentation

(a) Interpret the scenario

Different objects share similar intensity values, making simple thresholding ineffective.

(b) Key issues

- Intensity overlap.
- Noise.
- Irregular object shapes.
- Multiple regions.

(c) Apply advanced segmentation

Technique: Watershed Segmentation

Alternative:

- K-Means Clustering
- Region Growing

(d) Justification

Watershed Segmentation uses gradient information rather than intensity alone. It effectively separates touching or overlapping regions and performs well in complex images.

(e) Structured Explanation

Problem: Overlapping intensities

Technique: Watershed Segmentation

Result: Accurate separation of complex regions.

9. Broken Edges

(a) Interpret the scenario

After edge detection, object boundaries appear disconnected because weak edge pixels were removed or interrupted by noise.

(b) Key issues

- Discontinuous edges.
- Weak boundary detection.
- Missing edge pixels.
- Poor object representation.

(c) Apply a suitable method

Technique: Morphological Closing (Dilation followed by Erosion)

(d) Justification

Closing fills small gaps and connects nearby edge segments without significantly altering object shapes. This improves boundary continuity and prepares the image for further analysis.

(e) Structured Explanation

Problem: Broken boundaries

Technique: Morphological Closing

Result: Continuous and connected edges.

10. Combined Requirement

(a) Interpret the scenario

An industrial inspection system must remove noise while preserving important edges so that defects such as cracks, scratches, or dents can be detected accurately.

(b) Key issues

- Random image noise.
- Preservation of defect edges.
- High detection accuracy.
- Avoiding loss of fine details.

(c) Apply an appropriate sequence

Step 1: Apply Bilateral Filter for noise removal.

Step 2: Apply Canny Edge Detection.

(Optional Step 3: Morphological Closing to strengthen edge continuity.)

(d) Justification

The Bilateral Filter removes noise while preserving edges because it considers both spatial distance and intensity differences. Applying edge detection afterward ensures that important defect boundaries remain sharp and accurately detectable. If required, Morphological Closing can connect any broken edges, improving inspection reliability.

(e) Structured Explanation

Problem: Remove noise while preserving edges

Technique Sequence:

1. Bilateral Filtering
2. Canny Edge Detection
3. (Optional) Morphological Closing

Result: Clean image with preserved edges, enabling accurate defect detection.

11. Low Brightness Image

(a) Interpret the scenario and explain the cause of low brightness

A low-brightness image is captured under insufficient lighting conditions, making the entire image appear dark. This may occur due to poor illumination, incorrect camera exposure settings, or a low-sensitivity image sensor. As a result, important objects and details become difficult to observe.

(b) Identify key issues affecting image visibility

- Dark image with poor visibility.
- Loss of object details in shadow regions.
- Reduced contrast between objects and background.
- Difficulty in feature extraction and object recognition.
- Increased chances of noise in dark areas.

(c) Apply an appropriate enhancement technique

Technique: Gamma Correction ($\gamma < 1$) or Brightness Adjustment

Steps:

1. Read the input image.
2. Apply gamma correction with gamma less than 1 (e.g., $\gamma = 0.5$).
3. Increase pixel intensity values.
4. Obtain a brighter image while preserving details.

(d) Justify your choice

Gamma correction enhances dark regions without excessively brightening already visible areas. It produces natural-looking brightness improvements and preserves image quality better than simply adding a constant intensity value.

(e) Structured Explanation

Problem: Low brightness

Cause: Insufficient lighting or underexposure

Technique: Gamma Correction

Result: Improved brightness and visibility while maintaining image details.

12. Overexposed Image

(a) Interpret the scenario and explain overexposure

An overexposed image occurs when excessive light reaches the camera sensor. Many pixels become saturated (close to 255), causing bright regions to lose important details.

(b) Identify key issues affecting intensity representation

- Extremely bright image.
- Loss of details in highlighted regions.
- Pixel saturation.
- Reduced contrast.
- Poor feature visibility.

(c) Apply a suitable correction technique

Technique: Gamma Correction ($\gamma > 1$) or Intensity Normalization

(d) Justify your choice

Gamma correction with gamma greater than 1 reduces excessive brightness while preserving details in bright regions. Intensity normalization also redistributes pixel values to improve overall image balance.

(e) Structured Explanation

Problem: Overexposure

Cause: Excessive illumination

Technique: Gamma Correction ($\gamma > 1$)

Result: Balanced brightness and improved detail visibility.

13. Gaussian Noise

(a) Interpret the scenario and describe Gaussian noise

Gaussian noise is a random noise where pixel intensity values follow a normal (Gaussian) distribution. It usually appears due to sensor imperfections or electronic interference.

(b) Identify key issues affecting quality

- Random intensity fluctuations.
- Reduced smoothness.
- Blurred fine details.
- Lower image quality.
- Difficulty in edge detection.

(c) Apply an appropriate smoothing filter

Technique: Gaussian Filter

Steps:

1. Select Gaussian kernel.
2. Convolve the image with the kernel.
3. Smooth random noise while maintaining image appearance.

(d) Justify your method over alternatives

Gaussian filtering is specifically designed for Gaussian-distributed noise. It provides smooth noise reduction while preserving image continuity better than average filtering.

(e) Structured Explanation

Problem: Gaussian noise

Technique: Gaussian Filter

Result: Smooth image with reduced random noise.

14. Speckle Noise

(a) Interpret the scenario and explain speckle noise characteristics

Speckle noise is multiplicative noise commonly found in ultrasound, radar, and SAR images. It appears as grainy patterns that reduce image clarity.

(b) Identify key issues affecting clarity

- Grainy image appearance.
- Reduced contrast.
- Poor visibility of tissue boundaries.
- Difficult segmentation.
- Lower diagnostic accuracy.

(c) Apply a suitable filtering technique

Technique: Lee Filter (or Frost Filter)

Alternative: Adaptive Median Filter.

(d) Justify your approach

The Lee Filter is specially designed for multiplicative speckle noise. It smooths homogeneous regions while preserving important edges and structural information, making it ideal for medical ultrasound images.

(e) Structured Explanation

Problem: Speckle noise

Technique: Lee Filter

Result: Reduced speckle while preserving tissue boundaries.

15. Loss of Fine Details

(a) Interpret the scenario and explain detail loss

After image processing, fine structures disappear because aggressive filtering removes high-frequency components responsible for image details.

(b) Identify key issues related to filtering

- Over-smoothing.
- Loss of texture.
- Reduced edge sharpness.
- Poor feature visibility.
- Lower segmentation accuracy.

(c) Apply a suitable enhancement technique

Technique: Unsharp Masking

Alternative: High-Boost Filtering.

(d) Justify your method

Unsharp masking enhances high-frequency details by adding sharpened edge information back to the original image. It restores fine details without introducing excessive noise.

(e) Structured Explanation

Problem: Fine detail loss

Technique: Unsharp Masking

Result: Sharper image with restored details.

16. Edge Enhancement Requirement

(a) Interpret the scenario and explain the need for edge enhancement

Many computer vision applications require clear edges because edges represent object boundaries and important features used in recognition and analysis.

(b) Identify key issues in detecting edges

- Weak boundaries.
- Noise interference.
- Low contrast.
- Blurred edges.
- False edge detection.

(c) Apply an appropriate method

Technique: Laplacian Filter

Alternative: Sobel Operator.

(d) Justify your choice

The Laplacian operator highlights regions of rapid intensity change equally in all directions, making edges more prominent for feature extraction.

(e) Structured Explanation

Problem: Highlight edges

Technique: Laplacian Edge Enhancement

Result: Strong and clear object boundaries.

17. Background Noise Removal

(a) Interpret the scenario and explain noise impact

Background noise introduces unwanted pixel variations that interfere with object-background separation, making segmentation less accurate.

(b) Identify key issues affecting segmentation

- False object detection.
- Poor boundary accuracy.
- Increased segmentation errors.
- Reduced feature extraction performance.
- Misclassification.

(c) Apply a suitable preprocessing method

Technique: Bilateral Filter

Alternative: Gaussian Filter if edge preservation is less critical.

(d) Justify your approach

The Bilateral Filter removes noise while preserving important edges, improving segmentation accuracy without blurring object boundaries.

(e) Structured Explanation

Problem: Background noise

Technique: Bilateral Filtering

Result: Cleaner image with preserved edges and improved segmentation.

18. Threshold Selection Problem

(a) Interpret the scenario and explain the challenge

The image requires segmentation, but selecting one global threshold is difficult because illumination varies or object and background intensities overlap.

(b) Identify key issues in threshold selection

- Uneven illumination.
- Intensity overlap.

- Noise.
- Multiple object classes.
- Inconsistent segmentation.

(c) Apply a suitable thresholding method

Technique: Otsu's Thresholding

Alternative: Adaptive Thresholding.

(d) Justify your approach

Otsu's method automatically determines the optimal threshold by maximizing the variance between object and background classes, eliminating manual threshold selection.

(e) Structured Explanation

Problem: Difficult threshold selection

Technique: Otsu's Thresholding

Result: Automatic and accurate segmentation.

19. Texture-Based Segmentation

(a) Interpret the scenario and explain segmentation challenges

Different regions have similar intensity values but different texture patterns. Therefore, segmentation based only on pixel intensity cannot distinguish them.

(b) Identify key issues with intensity-based methods

- Similar intensity values.
- Texture differences ignored.
- Poor region separation.
- High segmentation error.
- Loss of structural information.

(c) Apply an appropriate segmentation approach

Technique: Texture-Based Segmentation using Gray Level Co-occurrence Matrix (GLCM) Features with K-Means Clustering

Alternative: Local Binary Pattern (LBP).

(d) Justify your method

GLCM extracts texture features such as contrast, correlation, energy, and homogeneity. Clustering these features allows accurate separation of regions even when intensity values are similar.

(e) Structured Explanation

Problem: Texture-based regions

Technique: GLCM + K-Means Clustering

Result: Accurate segmentation based on texture rather than intensity.

20. Multiple Object Detection

(a) Interpret the scenario and explain segmentation difficulty

The image contains several objects with nearly identical intensity values, making simple thresholding unable to distinguish between them.

(b) Identify key issues in distinguishing objects

- Similar intensity values.
- Touching or overlapping objects.
- Weak boundaries.
- Segmentation ambiguity.
- Incorrect object labeling.

(c) Apply a suitable segmentation technique

Technique: Watershed Segmentation

Alternative: Marker-Controlled Watershed.

(d) Justify your choice

Watershed segmentation uses gradient information and markers to separate touching objects, even when their intensity values are similar. It accurately identifies object boundaries and assigns each object a unique region.

(e) Structured Explanation

Problem: Multiple similar-intensity objects

Technique: Watershed Segmentation

Result: Accurate separation and identification of individual objects.

21. Edge Noise Problem

(a) Interpret the scenario and explain the issue

The image has undergone edge detection, but the detected edges contain unwanted noisy pixels. Noise creates false edges and interrupts the actual object boundaries, reducing edge detection accuracy.

(b) Identify key challenges in edge detection

- False edge detection due to noise.
- Broken or discontinuous edges.
- Difficulty identifying true object boundaries.
- Reduced accuracy in feature extraction.
- Poor segmentation performance.

(c) Apply a suitable solution

Technique: Gaussian Smoothing followed by Canny Edge Detection.

Steps:

1. Apply Gaussian Filter to remove noise.
2. Perform Canny Edge Detection.
3. Use hysteresis thresholding to retain only significant edges.

(d) Justify your method

The Gaussian filter removes random noise before edge detection. The Canny algorithm then accurately detects edges while suppressing false edges using non-maximum suppression and double thresholding.

(e) Structured Explanation

Problem: Noisy edges

Technique: Gaussian Filter + Canny Edge Detection

Result: Clean, continuous, and accurate edges.

22. Region Merging Requirement

(a) Interpret the scenario and explain the issue

After segmentation, the image contains many fragmented regions that actually belong to the same object. These regions need to be merged into meaningful segments.

(b) Identify key challenges in region segmentation

- Fragmented regions.
- Over-segmentation.
- Similar neighboring regions.
- Incorrect object representation.
- Increased computational complexity.

(c) Apply an appropriate technique

Technique: Region Merging Algorithm

Alternative: Morphological Closing.

(d) Justify your approach

Region merging combines adjacent regions with similar properties such as intensity, color, or texture, producing larger and more meaningful object regions.

(e) Structured Explanation

Problem: Fragmented regions

Technique: Region Merging

Result: Larger and accurate object regions.

23. Over-Segmentation

(a) Interpret the scenario and explain over-segmentation

Over-segmentation occurs when one object is divided into many small regions, making object analysis difficult.

(b) Identify key issues affecting segmentation

- Too many segmented regions.
- Increased processing time.
- Incorrect object representation.
- Difficulty in object recognition.
- Sensitive to noise.

(c) Apply a suitable correction method

Technique: Marker-Controlled Watershed Segmentation

Alternative: Region Merging.

(d) Justify your choice

Marker-controlled watershed reduces unnecessary region splitting by introducing predefined markers, leading to more accurate segmentation.

(e) Structured Explanation

Problem: Too many regions

Technique: Marker-Controlled Watershed

Result: Reduced over-segmentation and meaningful regions.

24. Under-Segmentation

(a) Interpret the scenario and explain under-segmentation

Under-segmentation occurs when multiple objects or regions are merged into one segment instead of being separated.

(b) Identify key issues affecting separation

- Objects merged together.
- Missing boundaries.
- Incorrect object counting.
- Reduced segmentation accuracy.
- Poor feature extraction.

(c) Apply a suitable solution

Technique: Watershed Segmentation with Distance Transform

(d) Justify your approach

Distance Transform identifies object centers and Watershed separates touching objects accurately, preventing incorrect merging.

(e) Structured Explanation

Problem: Merged objects

Technique: Distance Transform + Watershed

Result: Proper separation of individual objects.

25. Boundary Refinement

(a) Interpret the scenario and explain the issue

Detected boundaries are irregular, jagged, or incomplete, reducing segmentation quality.

(b) Identify key challenges in boundary detection

- Rough edges.
- Boundary gaps.
- Noise interference.
- Poor localization.
- Reduced measurement accuracy.

(c) Apply a suitable refinement method

Technique: Morphological Closing followed by Active Contour (Snake Model)

(d) Justify your choice

Morphological Closing connects broken boundaries, while Active Contours iteratively adjust the boundary to fit the true object shape accurately.

(e) Structured Explanation

Problem: Rough boundaries

Technique: Morphological Closing + Active Contour

Result: Smooth and accurate object boundaries.

26. Frequency-Based Enhancement

(a) Interpret the scenario and explain frequency-based processing

Instead of modifying pixels directly, the image is transformed into the frequency domain where low-frequency components represent smooth regions and high-frequency components represent edges and fine details.

(b) Identify key issues in spatial domain methods

- Difficult to isolate noise.
- Limited control over frequency components.
- May blur important details.
- Less effective for periodic noise.

(c) Apply an appropriate frequency domain technique

Technique: High-Pass Filtering using Fourier Transform.

Steps:

1. Apply FFT.
2. Apply High-Pass Filter.
3. Perform inverse FFT.

(d) Justify your choice

High-pass filtering enhances edges and fine details by emphasizing high-frequency components while suppressing low-frequency background information.

(e) Structured Explanation

Problem: Enhance image using frequency domain

Technique: FFT + High-Pass Filter

Result: Sharper image with enhanced details.

27. Mixed Noise Scenario

(a) Interpret the scenario and explain challenges

The image contains both Gaussian noise and impulse (salt-and-pepper) noise, making noise removal more challenging because one filter alone cannot effectively remove both types.

(b) Identify key issues affecting filtering

- Multiple noise types.
- Edge preservation.
- Fine detail protection.
- Increased filtering complexity.
- Risk of over-smoothing.

(c) Apply a suitable combined approach

Technique Sequence:

1. Median Filter (removes impulse noise).
2. Gaussian Filter (removes Gaussian noise).

(d) Justify your method

The Median Filter effectively removes salt-and-pepper noise without blurring edges. The Gaussian Filter then smooths the remaining Gaussian noise, producing a clean image with preserved structures.

(e) Structured Explanation

Problem: Mixed noise

Technique: Median Filter → Gaussian Filter

Result: Effective removal of both noise types.

28. Edge Preservation Filtering

(a) Interpret the scenario and explain the requirement

The application requires removing image noise while maintaining sharp object boundaries for accurate analysis.

(b) Identify key issues in filtering

- Noise reduction.
- Edge preservation.
- Avoiding excessive smoothing.
- Maintaining feature accuracy.

(c) Apply a suitable technique

Technique: Bilateral Filter

Alternative: Guided Filter.

(d) Justify your approach

The Bilateral Filter considers both spatial distance and pixel intensity differences. Therefore, it smooths homogeneous regions while preserving important edges.

(e) Structured Explanation

Problem: Noise removal with edge preservation

Technique: Bilateral Filter

Result: Clean image with preserved edges.

29. Region Growing Failure

(a) Interpret the scenario and explain the issue

Region Growing fails because incorrect seed points are selected. As a result, regions expand into wrong areas or fail to capture the complete object.

(b) Identify key challenges in region growing

- Incorrect seed selection.
- Uneven intensity.
- Noise.
- Poor region expansion.
- Incomplete segmentation.

(c) Apply a suitable correction

Technique: Automatic Seed Selection with Preprocessing

Steps:

1. Remove noise.
2. Detect homogeneous regions.
3. Automatically select seeds.
4. Perform Region Growing.

(d) Justify your method

Automatic seed selection minimizes human error and ensures that region growth starts from representative pixels, leading to more accurate segmentation.

(e) Structured Explanation

Problem: Incorrect seed selection

Technique: Automatic Seed Selection + Region Growing

Result: Accurate region segmentation.

30. Industrial Inspection Enhancement

(a) Interpret the scenario and define requirements

An industrial inspection system must accurately detect surface defects such as cracks, scratches, dents, or holes in noisy images while minimizing false detections.

(b) Identify key issues in detection

- Image noise.
- Low contrast defects.
- Weak edges.
- False positives.
- Accurate defect localization.

(c) Apply an appropriate enhancement and segmentation pipeline

Processing Pipeline:

1. Bilateral Filter (Noise Removal)
2. CLAHE (Contrast Enhancement)
3. Canny Edge Detection
4. Morphological Closing
5. Watershed Segmentation
6. Defect Detection and Analysis

(d) Justify your sequence

- **Bilateral Filter** removes noise while preserving edges.
- **CLAHE** improves local contrast, making defects more visible.
- **Canny Edge Detection** identifies accurate defect boundaries.
- **Morphological Closing** connects broken edges.
- **Watershed Segmentation** separates individual defect regions.
- This sequence provides accurate and reliable defect detection.

(e) Structured Explanation

Problem: Detect defects in noisy industrial images

Pipeline:

1. Bilateral Filtering

2. CLAHE
3. Canny Edge Detection
4. Morphological Closing
5. Watershed Segmentation

Result: Accurate defect detection with reduced noise and preserved edges.